

Chapter 1 introduced *Mathematica* by showing you how to use some of its more common features. This chapter looks at *Mathematica* in a different way.

Instead of discussing individual features, it describes the overall framework of *Mathematica*. You will be able to do many calculations in *Mathematica* without understanding the framework. However, if you want to use *Mathematica* to its fullest, you will need to know the framework. In addition, knowing the framework will help you remember how all the individual features of *Mathematica* work. In fact, you will probably often be able to “predict” how features work, without even reading the description.

This chapter assumes no advanced knowledge of computing. If you are an expert on computer languages, you may find it useful to look first at the Reference Guide. You should still read this chapter, though, to understand *why Mathematica* is constructed the way it is.

About the illustration overleaf:

This is a picture of a regular dodecahedron. The package `Polyhedra.m` includes the function `Dodecahedron`, which gives a list of graphics primitives representing the faces of a dodecahedron. The complete *Mathematica* command used to produce the picture was `Show[ Graphics3D[Dodecahedron[ ]] ]`.